

Understanding World Wide Web

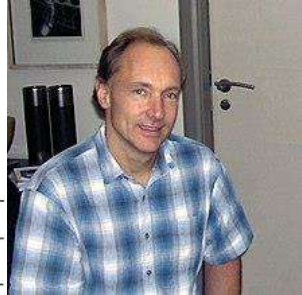
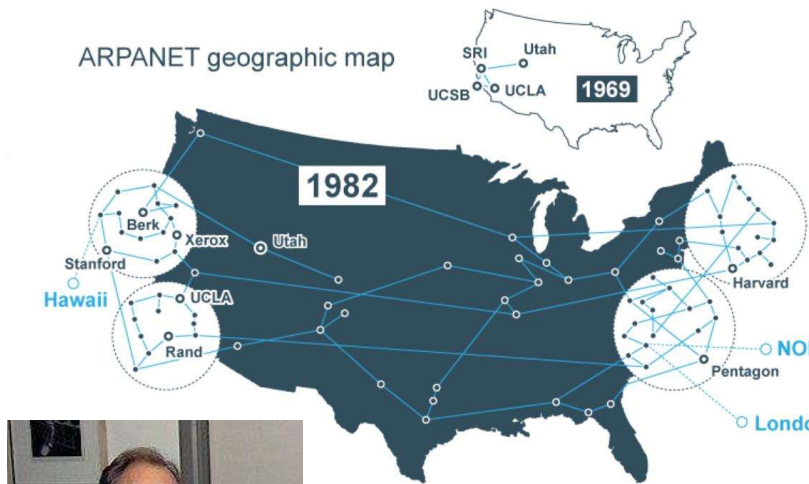
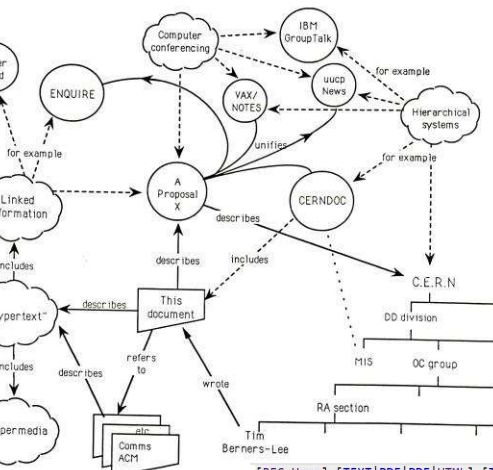
Software System Development
Monsoon 2026

Information Management: A Proposal

Abstract

This proposal concerns the management of general information about accelerators and experiments at CERN. It discusses the problems of loss of information about complex evolving systems and derives a solution based on a distributed hypertext system.

Keywords: Hypertext, Computer conferencing, Document retrieval, Information management, Project control



Dot-decimal notation
Binary numbering
Octet's order

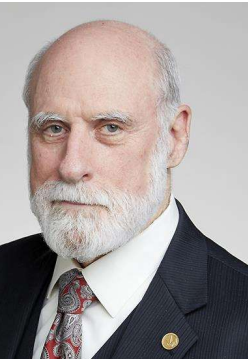
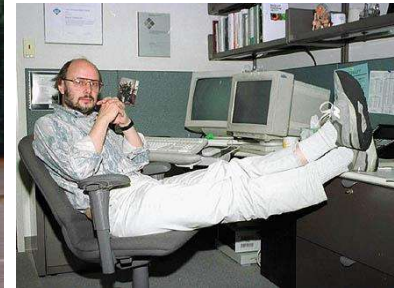
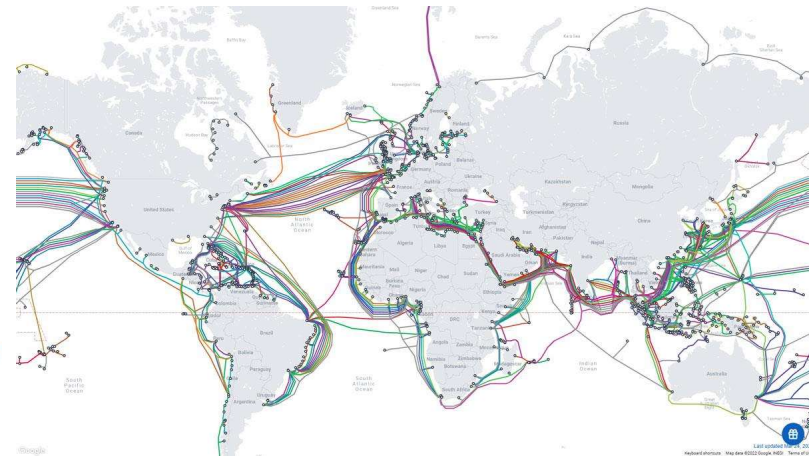
Most significant bit

172 . 16 . 254 . 1

10101100.00010000.11111110.00000001

1st 2nd 3rd 4th

32-bit (4 bytes)



[RFC Home] [TEXT] [PDF] [PDF] [HTML] [Tracker] [IPR] [Errata] [Info page]

INTERNET STANDARD
Errata Exist
Vint Cerf
UCLA
Request for Comments: 20
October 16, 1969

ASCII format for Network Interchange

For concreteness, we suggest the use of standard 7-bit ASCII embedded in an 8 bit byte whose high order bit is always 0. This leads to the standard code given on the attached page, copies from USAS X3, 4-1968. This code will be used over HOST-HOST primary connections. Break characters will be defined by the receiving remote host, e.g. SRI uses "." (ASCII X'2E' or 2/14) as the end-of-line character, where as UCLA uses X'0D' or 0/13 (carriage return).

USA Standard Code for Information Interchange

1. Scope

This coded character set is to be used for the general interchange of information among information processing systems, communication systems, and associated equipment.



<https://www.rfc-editor.org>



<https://www.whois.com>

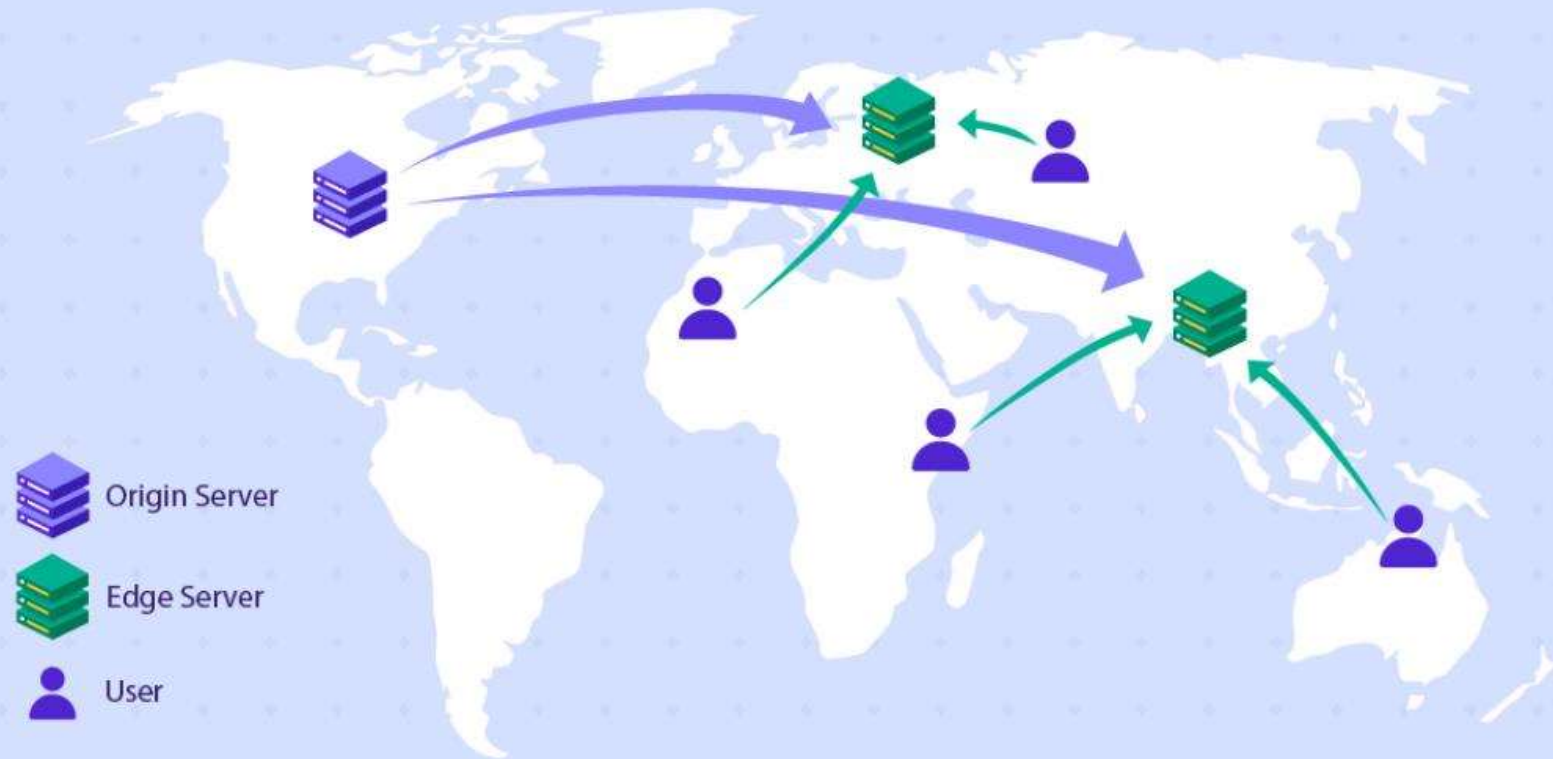
<https://www.datacentermap.com>



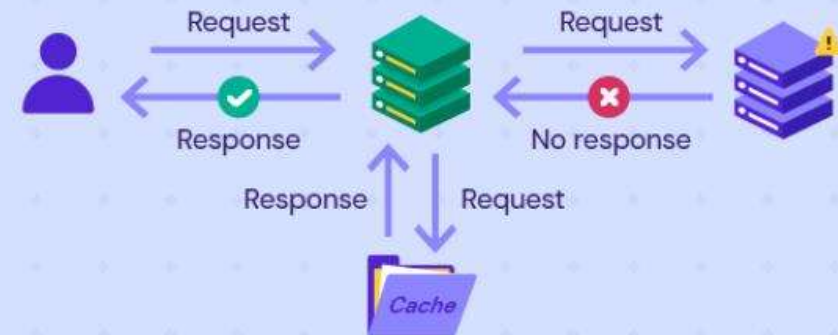
Content Delivery Network



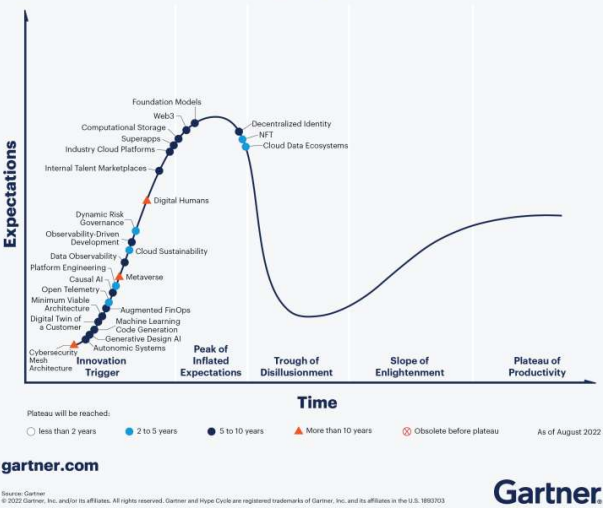
How Does a CDN Work?



When a user visits a static website for the first time, a CDN server will transfer the files from the origin server. Note that you can access the content even when the origin server is not working since a CDN server typically stores and serves the static cached data.



Hype Cycle for Emerging Tech, 2022



Web 1.0

READ-ONLY
Almost No-Interaction
Company-Centered

Static Web Pages,
Encyclopedia, Phonebook,
Static News, Information

Web 2.0

READ+WRITE
Shared-Dynamic
Community-Centered

Email, Wikipedia, Social
Network, Stock Market, E-
News, Video Websites, Mobile

Web 3.0

READ-WRITE-OWN
AI-driven Centered
Individual-Centered

Semantic Web, VR/AR/MR,
MetaVerse, DeCentralized
Social Networks, NFTs

Web 4.0

??

HTML – Hyper Text Markup Language



Instruction:
a paragraph
with bold text



Displays a paragraph
with bold text

```
<!DOCTYPE html>
<html>
  <head>
    <title> My Home Page
  </title>
  </head>
  <body>
    <p>About myself.</p>
  </body>
</html>
```

- HTML declaration
- HTML opening tag
- Page header opening tag
- Page title opening tag
- Page title closing tag
- Page header closing tag
- Body opening tag
- Content (inside body)
- Body closing tag
- HTML closing tag



Address bar

Browser



HTTP

iiit.ac.in



Give me the
/admin page

Give me page
<https://intranet.iiit.ac.in/admin>

HTML Validator: <http://validator.w3.org>



HTTP

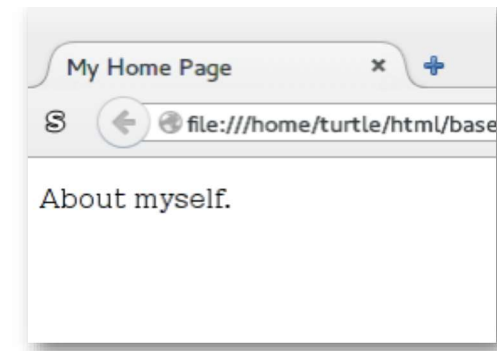
/admin
HTML page



Rendering of
/admin page



```
<!DOCTYPE html>
<html>
  <head>
    <title> My Home Page
  </title>
  </head>
  <body>
    <p>About myself.</p>
  </body>
</html>
```



- **HTML** stands for **H**yper**T**ext **M**arkup **L**anguage
- Standard markup language for creating web pages
- Describes the structure of a webpage semantically
- Cornerstone technology of the World Wide Web, alongside CSS and JavaScript

- An HTML document has a nested structure defined by tags
- The basic skeleton includes:
 - `<!DOCTYPE html>` declaration
 - `<html>` element that wraps all content
 - `<head>` for metadata and scripts/styles
 - `<body>` for visible content



The content inside the `<body>` section will be displayed in a browser. The content inside the `<title>` element will be shown in the browser's title bar or in the page's tab.

- An HTML element consists of:
 - Start tag: <tagname>
 - Content: Text and other elements
 - End tag: </tagname>
- Some elements are self-closing (e.g., ,
)

Examples

```
<h1>My First Heading</h1>
<p>My first paragraph.</p>
```

- Modern browsers have developer tools for inspecting elements
- Access methods:
 - Right-click + "Inspect" or "Inspect Element"
 - Keyboard: F12 or Ctrl+Shift+I (Cmd+Option+I on Mac)
- Uses:
 - View HTML structure
 - Edit CSS styles in real-time
 - Debug JavaScript
 - Test responsive designs

- 1 **Create a new text file** using any text editor
- 2 **Write your HTML code**

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Web Page</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is my first web page.</p>
</body>
</html>
```

- 3 **Save with .html extension** (e.g., index.html)
- 4 **Open in a web browser** (double-click the file)

HTML Attributes

- Provide additional information about elements
- Always specified in the start tag
- Format: `<element attribute="value">`
- Common attributes:
 - id - Unique identifier
 - class - Group elements for styling
 - style - Inline CSS styles
 - src - Source for embedded content
 - href - Hyperlink target

Example

```
<a href="https://example.com" target="_blank">
  Visit Example
</a>
```

Paragraphs & Headings

```
<p>This is a paragraph.</p>
```

```
<h1>Main Heading</h1>
```

```
<h2>Subheading</h2>
```

```
<h3>Section heading</h3>
```

Links (Anchors)

```
<!-- External link -->
```

```
<a href="https://example.com">
  Visit Example
</a>
```

```
<!-- Page link -->
```

```
<a href="about.html">About</a>
```

```
<!-- Section link -->
```

```
<a href="#section1">Section</a>
```


Images

```



```

Line Breaks & Dividers

```
<p>Line 1<br>Line 2</p>

<p>Above the line</p>
<hr>
<p>Below the line</p>
```

Title and Favicon

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>
  <link rel="icon"
        href="favicon.ico"
        type="image/x-icon">
</head>
<body>
  <!-- Content -->
</body>
</html>
```

HTML Comments

```
<!-- This is a comment -->

<!--
  Multi-line
  comment
-->
```

Unordered List

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

Ordered List

```
<ol>
  <li>First step</li>
  <li>Second step</li>
  <li>Third step</li>
</ol>
```

Description List

```
<dl>
  <dt>Coffee</dt>
  <dd>Black hot drink</dd>
  <dt>Milk</dt>
  <dd>White cold drink</dd>
</dl>
```

`<b>Bold text</b>`
`<strong>Important text</strong>`
`<i>Italic text</i>`
`<em>Emphasized text</em>`
`<mark>Highlighted text</mark>`

`<small>Smaller text</small>`
`<del>Deleted text</del>`
`<ins>Inserted text</ins>`
`<sub>Subscript text</sub>`
`<sup>Superscript text</sup>`

Example Usage

```
<p>Water is H<sub>2</sub>O. The result is 5<sup>2</sup> = 25.</p>
<p>Price: <del>$50</del> <ins>$40</ins> <strong>Sale!</strong></p>
```

Block-level Elements

- Start on a new line
- Take full width available
- Examples:
 - `<div>`
 - `<h1>-<h6>`
 - `<p>`
 - `<ul>`, `<ol>`
 - `<table>`

Inline Elements

- Do not start on a new line
- Take only necessary width
- Examples:
 - `<span>`
 - `<a>`
 - `<img>`
 - `<strong>`, `<em>`
 - `<b>`, `<i>`

Div Element

- Block-level container
 - Groups content for styling
 - Creates natural breakpoints
- ```
<div class="container">
 <h2>Section Title</h2>
 <p>Paragraph text...</p>
</div>
```

### Span Element

- Inline container
  - Style parts of text
  - No visual change by default
- ```
<p>
  This is <span class="highlight">
    important</span> text.
</p>
```

HTML Classes

- Group similar elements
- Reusable across many elements
- Element can have multiple classes

```
<div class="box highlight">  
    Content  
</div>
```

```
<p class="highlight">  
    Highlighted paragraph  
</p>
```

HTML IDs

- Unique identifier
- Only one per page
- Used for JavaScript access
- Used as anchor links

```
<div id="header">  
    Header content  
</div>
```

```
<a href="#section1">  
    Jump to Section 1  
</a>  
<div id="section1">  
    Section content  
</div>
```